

Spontaneous Imagery Scanning in Mental Extrapolation

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We report an experiment that suggests a functional application of mental-image scanning. After subjects inspected a simple dot pattern, the pattern was removed, and they were then shown an arrow at an unexpected location. Their task was to judge as quickly as possible whether the arrow pointed at any of the dots in the previously observed pattern. Although the subjects were never instructed to form or scan mental images, most of them reported having done so in order to make their judgments, and their reaction times were directly proportional to the distances separating the dots and the arrows, as typically found in image-scanning experiments. Imagery scanning may therefore serve a useful function when one must judge spatial relations between the positions of remembered objects and newly specified locations.

Several recent experiments have shown that mental images can be scanned with a duration that increases linearly with increasing distance between the source and destination of scanning (Kosslyn, 1973; Kosslyn, Ball, & Reiser, 1978). These experiments have provided evidence that images can accurately depict relative distances among physical objects (see Kosslyn, Pinker, Smith, & Schwartz, 1979a) and have also provided a useful measure of the spatial perspective properties of images (see Pinker, 1980; Pinker & Finke, 1980). However, the precise function of image scanning is unclear.

Although the ostensible purpose of image scanning in the Kosslyn et al. (1978) experiments was to determine whether or not a named object was contained within a collection of objects previously observed, such decisions can usually be made more rapidly by not forming mental images (e.g., see Kosslyn, 1976; Kosslyn et al. 1978). Moreover, in a typical image-scanning experiment, subjects first learn the relative positions of a set of objects and are then given explicit instructions to scan between the locations of the objects as they are "seen" in their mental images. Critics of this work

have argued that this experimental procedure taps not a functional capacity of the mind, but knowledge of the laws governing physical motion. Indeed, other recent studies have shown that subjects can infer that scanning times in such experiments ought to be linearly related to the distances scanned (e.g., Mitchell & Richman, 1980). Although we do not believe that the results of all image-scanning experiments can properly be attributed to such "demand characteristics" (see Finke, 1980; Kosslyn, Pinker, Smith, & Schwartz, 1979b), it would be important to find tasks in which subjects form and scan mental images without having to be told to do so.

At first we conceived of the following experimental task as one in which imagery scanning might have a demonstrable, practical application: Subjects first learn the relative positions of a set of distinct circular objects separated by various distances and are then shown one of the objects in a tachistoscope followed by an arrow pointing outward from the center of the object. They are instructed to respond "yes" if the arrow is pointing at any other object of the set, and to respond "no" otherwise. We thought that in order to make these judgments the subjects would form and scan a mental image of the configuration of objects, extrapolating along the particular direction indicated by the arrow in order to "see" whether or not another object would intercept the extrapolated line. This would be suggested by a linear relationship between reaction time and distance between the objects.

To our disappointment, when pilot subjects were actually tested under these conditions, we

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found no relationship between distance and reaction time. We then repeated the above experiment, trying a number of simple variations such as changing the size or number of the objects slightly; but in each case reaction time and interobject distance were essentially uncorrelated. We later learned that analogous results were obtained by Wilton (1979) and Bannon (1980). These results suggested that people encode information about relative directions between pairs of objects at the time they attempt to learn their locations, making it unnecessary to form and to scan mental images when they later judge whether test directions connect the objects. Apparently, people can directly compare a test direction from an object and the remembered direction between that object and another in much the same way that one would compare the relative directions of two vectors pointing outward from the same location.¹

However, this method of comparison could not be used if judgments of direction were required from an unexpected location relative to a set of remembered objects. Suppose, for example, that subjects were to observe first a collection of objects, and then an arrow pointing from an arbitrary location in an otherwise blank field. Again, their task is to respond "yes" only if the arrow is pointing to any of the locations where the objects had previously appeared. In this case the subjects could have no explicit memory for the correct directions to the objects from that arbitrary location; hence, they would more likely need to scan along an extrapolated line in an image of the configuration in order to perform the task. That is, if image scanning has any functional value, it would surely be to facilitate judgments of spatial relations among perceived and remembered objects that could not have been explicitly encoded at an earlier time. The experiment we presently report supports this proposal.

Method

Subjects

Twelve students at Cornell University, six males and six females, served as paid volunteer subjects. Prior to participation they were informed only that the experiment would be on "visual memory."

Procedure

The subjects were tested individually. They were seated in front of a Gerbrands two-channel tachistoscope and observed stimuli presented within a square field $19^\circ \times 19^\circ$ at a viewing distance of 57.5 cm. At the beginning of each trial, the subjects freely inspected a dot pattern that

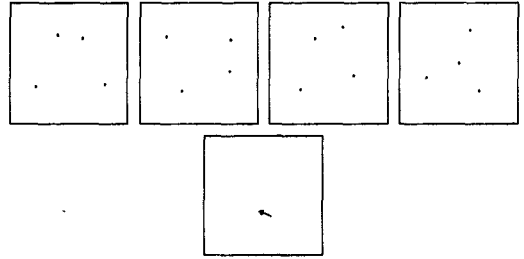


Figure 1. Dot patterns used for the experimental trials (above) and an example of an arrow stimulus (below) on which judgments of direction were then made. (For purposes of illustration the arrows and dots have been drawn slightly larger than their actual size relative to the size of the fields.)

was presented for 5 sec, with instructions to remember the positions of the dots. Each of the patterns contained four small black dots, 6 mm in diameter, on a white background; the four patterns used for the experimental trials are shown in the upper part of Figure 1. A blank field was then presented for 1 sec, followed by a field that contained a single black arrow, 13 mm in length, that was presented for 4 sec. An example of one of the arrow stimuli is shown in the lower part of Figure 1. The locations of dots and arrows were restricted to a circular region extending 7 cm in radius from the center of the field. The white backgrounds for the arrow and dot fields were matched for luminance.

The subjects' task was to respond "yes" if the arrow on a particular trial was pointing to any of the dots that they had just observed, and to respond "no" if the arrow was not. They were instructed to do so as quickly and as accurately as they could as soon as the arrow was presented by pushing the appropriate button on a response box. The subjects were informed that this judgment task would not be difficult, because the arrows would either be pointing directly at a dot or clearly miss all of the dots. In fact, on no trials did an arrow point closer than 40° to a dot without pointing directly at it; thus, there were no trials on which an arrow "barely missed" a dot. To further avoid confusion none of the arrows were presented at locations at which dots had just been presented.

Once these instructions had been given, the subjects were shown sample stimuli and practiced pushing the "yes" and "no" buttons on the response box. Four practice trials were then conducted, followed by the 32 experimental trials.

¹ Indeed, even in Kosslyn et al.'s (1978) experiments, there was probably some explicit encoding of relative direction, allowing subjects to know in advance in what direction to begin scanning.

Each of the four patterns shown in Figure 1 was presented on eight consecutive trials, so that the effect of pattern familiarity could be separately examined from that of overall practice. On five of the trials, the arrow pointed at one of the dots on the pattern, and on the other three trials, the arrow completely missed all of the dots. For those trials on which the arrow did point at a dot, the distance between the arrow and the dot was varied, being either 4, 6, 8, 10, or 12 cm (to reduce the salience of the distance manipulation a separation distance of 2 cm was not included). Each distance occurred only once for each pattern. For those trials on which the arrow missed the dots, one of the arrows was repeated. The locations and orientations of arrows for each type of response were equally distributed across the field.

The association between arrow distance and dot location was randomized for each of the patterns. Because there were five levels of arrow distance and four dots on each pattern, one of the dots on every pattern was pointed at twice (though not from the same distance). The order in which the patterns were presented was randomized across subjects, as was the order in which the arrow stimuli were presented, with the constraint that no more than three "yes" trials or two "no" trials occur consecutively. The experimental trials were conducted without interruption, and the subjects were given feedback regarding their accuracy. Errors were relatively infrequent, occurring on 4.6% of the trials. At the conclusion of the experiment, each subject was asked to describe how he or she performed the task.

Results

The mean reaction time for correctly detecting that an arrow was pointing at a previous dot location increased with the distance between the arrow and the dot, $F(4, 40) = 9.79, p < .001$. The linear trend for this effect was highly significant, $F(1, 40) = 34.60, p < .001$, whereas all higher order trends failed to reach significance, $F(3, 40) = 1.57, p > .10$; this was also indicated by the overall correlation between reaction time and distance ($r = .94$). In addition, as shown in Figure 2, increasing familiarity with the patterns tended to strengthen this linear relationship. Reaction time and distance were related linearly for each of the four individual patterns as well, with correlations ranging from .71 to .89.

Even though the subjects were never told to form images, 6 of the 12 reported, when interviewed at the end of the experiment, that they had found it necessary both to form mental images of the patterns and to scan their images along the directions indicated by the arrows in order to

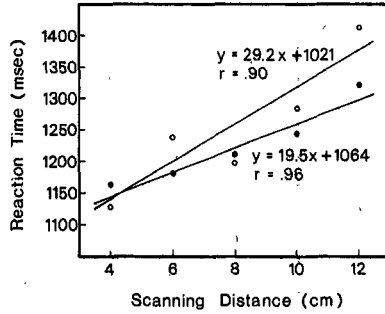


Figure 2. Mean reaction times for increasing distances separating the dot and arrow locations. (The open circles correspond to reaction times for the first half of all trials for each pattern, whereas the closed circles correspond to reaction times for the second half of all trials [$N = 12$].)

make their judgments. Five of the remaining six subjects reported that they had remembered using images of the patterns in some way, whereas the last subject reported not having used images.² These reports, together with the reaction time data, suggest that this kind of task is typically performed by scanning along a straight line in a mental image.³ Of further significance, the scanning rate of 19.5 msec/cm obtained when subjects became familiar with the patterns (see Figure 2) converts to a rate of 19.8 msec/degree of visual angle, which is very close to the scanning rate of 17 msec/degree obtained in experiments where subjects are given explicit instructions to form and scan mental images (Kosslyn, 1978, Experiment 5).⁴

Analyses on sex differences showed that the

² This last subject, incidentally, also showed the smallest individual correlation between reaction time and distance among the 12 subjects.

³ We later attempted, out of curiosity, to run a colleague in the same experiment who professed to have no mental imagery whatsoever. Whereas all the previous subjects reported that the task was not difficult, this particular subject reported being completely unable to make any of the direction judgments at all, further strengthening our claim that image scanning is important in performing the task.

⁴ Other estimates of mental scanning rate are in the same range. Pinker (1980) found rates of 13.0 msec/degree for scanning across images of three-dimensional configurations, and 18.6 msec/degree for scanning across those configurations when they were visible. Note that in contrast to these roughly equivalent estimates, Mitchell and Richman's (1980) subjects estimated that they would take 240 msec for each cm scanned, corresponding to a scanning rate of about 180 msec/degree if we assume that the map they used was held at a normal reading distance.

rate at which reaction time increased with increasing distance was smaller for males than it was for females, $F(4, 40) = 2.93, p < .05$. This last finding should be interpreted with some caution, however, given the general inconsistency of findings relating sex differences to differences in performance on mental imagery tasks (e.g., see Cooper & Regan, in press).

Discussion

In this section we consider various implications and alternative accounts of these results. First, in contrast to the majority of previous studies on imagery scanning, our subjects could not simply have retrieved explicit encodings of interpoint distances and then delayed their responses by proportional amounts. This is because information about each scanning distance was never available to the subjects until the arrow was actually presented, at which time they had to make an immediate judgment about its direction. They were unlikely to have had sufficient time to encode explicitly information about the distance between the arrow and the dot until after they had made their judgments. More important, because there were no imagery instructions of any sort, they would have had no motive for manipulating their response times even if they had encoded the arrow-dot distances. They had no reason to expect that the experimenter was looking for a distance-time relation (cf. Mitchell & Richman, 1980), nor should the instructions have reminded them of some physical or perceptual chain of events that they would then take it upon themselves to simulate mentally (cf. Pylyshyn, 1981). Indeed, all of the subjects reported that they thought the purpose of the experiment had been to test their visual memory, and several reported that they didn't even realize that arrow distance had been manipulated until this was later revealed to them at the experiment's conclusion.

One possible problem with the experiment is that arrows corresponding to greater scanning distances might have been located farther away from the center of the field than were arrows corresponding to shorter distances. If more time is required to see arrows that are farther away from the center of the field (either because of the duration of eye movements from the average point of fixation or the longer detection times for more retinally eccentric stimuli), this might account for the increase in reaction time with increasing arrow-dot distance. To address this possibility we measured the distance of each arrow from the center of the field and then computed the correlation between reaction time and arrow-dot distance, partialing out the contribution of this vari-

able. The resulting partial correlation increased to .97, whereas the relationship between dot distance and field location was quadratic, having a relatively low correlation of .31. These two variables were therefore roughly independent and together accounted for 96% of the variation in mean reaction time ($R = .98$).

Another possible problem is that subjects could have used visual afterimages of the previously observed patterns when making their judgments instead of using mental images *per se*. This is unlikely for the following reasons: First, the subjects were free to move their eyes when inspecting the patterns, which would serve to weaken any such afterimages. Second, a delay period was included between the offset of the pattern and the onset of the arrow, which allowed "iconic" types of images to fade and thereby discouraged reliance on such residual visual effects. Third, the more persistent negative afterimages would move with any eye movements directed to the arrow locations; in fact, the one subject who reported attempting to use pattern afterimages to perform the task said that he recognized this and quickly abandoned the strategy. Several other subjects also noticed such afterimages and reported them to be distracting, not helpful.

It is also conceivable that our subjects might have moved their eyes from the arrow locations to the remembered dot locations in order to make their judgments instead of mentally scanning between those locations. However, even if saccadic eye movements of this type had occurred on every trial, they would not explain the present rate at which reaction time increased with increasing scanning distance. Such eye movements are reported to have a velocity of 2 msec/degree (see Alpern, 1972), which is nearly 10 times as rapid as the scanning rates obtained here or in previous imagery-scanning studies (see Kosslyn, 1978; Pinker, 1980).⁵

In addition, one could argue that our subjects were just being more conservative when judging directions that involved greater distances between the dots and the arrows. This criticism would be especially appropriate had trials been included in which arrows barely missed the edge of a distant dot. Because our subjects knew in advance that not one of their judgments would involve making difficult discriminations of this type between extrapolated directions and dot locations, we believe that this last criticism also would not apply to the conditions of the present study. In any case, error

⁵ Pinker (1980) obtained eye-movement rates of 5.0 msec/degree for three-dimensional displays, also many times faster than the image-scanning rates measured here and elsewhere.

rates were positively correlated with distance ($r = .59$), so this possibility may be readily dismissed.

One could point out, however, that a positive relationship between errors and distance may reflect a response selection effect. If, for example, subjects with faster overall reaction times produced more errors on the longer distance trials than did subjects with slower reaction times, this difference might account for the average increase in reaction time with distance. There are several reasons why a response-selection effect could not have been responsible for our results. First, the subjects' individual reaction times for each arrow-dot distance were averaged before being entered into the statistical analyses so that fast and slow subjects contributed an equal number of data points. Second, the actual number of errors for levels of increasing distance were, respectively, 0, 1, 6, 2, and 4—showing that the most errors were made on the intermediate-distance trials. Third, the four subjects who did make errors on trials for the two longest distances showed mean reaction times that were slightly slower than those of the other eight subjects. Finally, four of the subjects made no errors on any of the trials, and they each showed an increase in reaction time with distance.

We should also consider several limitations of the present study. For instance, our findings do not distinguish among the possible types of mental images that could be used in performing the task. Because the dot patterns were always displayed before the arrows, our conclusions may be restricted to the kinds of mental images that are retained in short-term visual memory. Additional experiments will be needed to establish whether these findings can be extended to cases where images are formed entirely from long-term memory. In this regard the apparent "flattening" of our reaction time function with increasing pattern familiarity might suggest that the distance effect would disappear altogether when there is extensive learning of the patterns. However, because this flattening was accompanied by an increase in linearity, we do not think that further increases in familiarity would substantially alter the general shape of this function. And although we would certainly agree that the presence of a linear effect of distance on reaction time does not necessarily demonstrate mental-image scanning, we feel that the particular rate of increase in reaction time, together with our subjects' own accounts of how they performed the task, are most consistent with an image-scanning interpretation.

To conclude, the results of this study suggest that a possible functional application of mental-image scanning is to facilitate judgments of spa-

tial relations between remembered objects and arbitrary unexpected locations. Specifically, imagery scanning may be most useful when it is necessary to mentally extrapolate along a particular direction connecting objects that have not previously been observed together, in order to verify that one lies in a particular direction with respect to the other. In contrast, when objects are observed together, directions between the objects can be explicitly remembered, and imagery scanning need not be used in making judgments about directions. We believe that is why distance effects were absent in the studies cited in the Introduction but were obtained in the present study. In general, these interpretations are consistent with the view that attributes to mental imagery the special function of facilitating incidental recall and inferential reasoning, that is, the extraction of information not explicitly encoded at the time of initial learning (e.g., Sheehan, 1972; Waltz, 1979; Pinker, Note 1).

Reference Note

1. Pinker, S. *Mental imagery and the visual world* (Center for Cognitive Science Occasional Paper No. 4). Unpublished manuscript, Massachusetts Institute of Technology, 1980.

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